

CoRoT-2b

R-2257

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_120608 · created 2026-08-16T00:47:43+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456086.8559 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.012
Transit depth (Rp/Rs)^2	3.66 +/- 0.45 [%]
Orbital Inclination (inc)	89.85 +/- 0.85
Airmass coefficient 1 (a1)	0.994 +/- 0.02
Airmass coefficient 2 (a2)	0.005 +/- 0.038
Scatter in the residuals of the lightcurve fit is	2.04 %
Best Comparison Star	#3 - [201, 179]
Optimal Method	PSF photometry
Transit Duration (day)	0.0935 +/- 0.0012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.191 ± 0.012 vs 0.1667 (deviation 2.02σ)
Duration fit vs published	0.0935 d vs 0.095 d (deviation 1.25σ)
Mid-time O-C	-0.6 min
Depth SNR	15.9
Residual scatter	2.04 %

Light curve

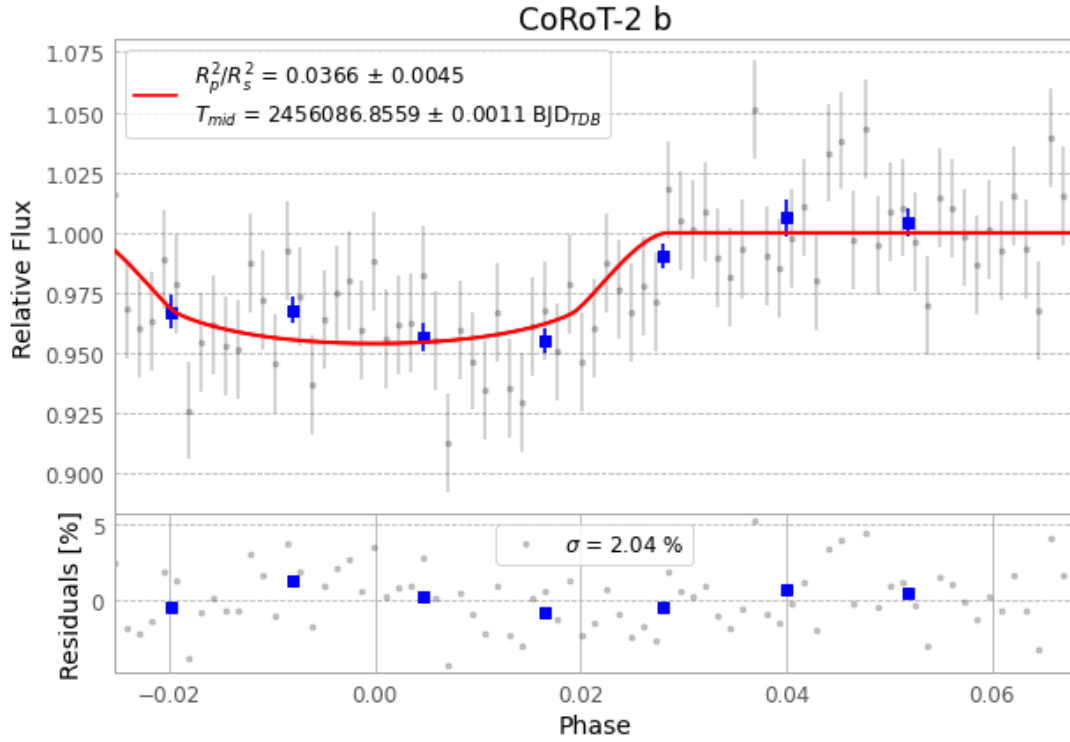


Figure 1 — FinalLightCurve_CoRoT-2 b_2012-06-08.png (SHA-256 fcd9b23f6d184f3e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [337, 240], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c7d31235425b4f15...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

79 FITS frames (52 MB), CoRoT-2120608072112.FITS → CoRoT-2120608111513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-120608.log (af464a80562b...), FinalLightCurve_CoRoT-2 b_2012-06-08.png (fcd9b23f6d18...), FinalLightCurve_CoRoT-2 b_2012-06-08.csv (8a7087548b92...)

Compute resources

Wall time 128.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 00:47Z.